PEDAVER
THE TRANSFORMATIVE PRODUCER

Knowledge Paper 198

PQNK - Natural Ecosystem Science for Production Agriculture

Abundance of nutritious food
PQNK®
The science of natural farming

SOIL | PLANTS | PEOPLE | A HEALTHY PLANET

The Seed Is Potential. The Soil Is the Production System.

Rethinking High-Yielding Varieties, Heirloom Seeds, Hybrids and GM Crops Through the PQNK Lens

Genetics determines what is possible. A living soil determines what is realised.



> *The future of agriculture should not require choosing between science and nature, or between modern genetics and traditional genetics. The real task is to place genetics, old or new, back inside the production system that made plant life possible in the first place.*
>
> *— PQNK*

1. Introduction

A simple question raised during discussions among farmers leads directly to one of the least examined assumptions of modern agriculture:

"Why are Green Revolution varieties described as 'fertiliser-responsive'? Does that mean these plants require synthetic fertiliser? Would the same seed respond differently if planted in a fully functioning living soil?"

The seed carries genetic potential. The environment determines how that potential is expressed. The soil ecosystem is a major part of that environment.

Key Messages

- **Genetics is potential, not the production system.**
- **Soil structure unlocks potential.**
- **All varieties remain plants governed by biology.**
- **Save natural, traditional and locally adapted seed.**
- **Seed sovereignty is food sovereignty.**
- **In maize, pollen-mediated gene flow can move GM traits into non-GM seed.**

2. A Seed Is Stored Genetic Potential

Every seed contains inherited biological instructions from which the next plant develops. Natural evolution modifies those instructions through variation, reproduction and selection within particular environments. Human breeding accelerates selection toward characteristics that people value.

A wheat seed may possess genes for grain production, stature, disease resistance, root architecture, grain quality or adaptation to stress. Whether those characteristics are fully expressed depends upon the environment in which the plant grows.

The genotype establishes possibilities. The growing environment determines expression.

The seed carries genetic potential. The environment determines expression.

3. Soil Has the Same Relationship Between Inheritance and Function

PQNK's work on soil texture and soil structure provides a useful parallel. Soil texture is largely inherited: sand, silt and clay are the mineral materials with which a soil begins. Structure is dynamic. Roots, fungal hyphae, microbial products, organic residues, pores, water and air organise those particles into a functioning living system.

PQNK therefore does not attempt to turn sand into clay or clay into sand. It restores the biological structure through which the soil functions. Under improved structure, sandy soils can retain more water and nutrients, while clay soils can develop greater porosity, infiltration and root penetration.

Genetics is to the plant what texture is to the soil: inherited potential. The production environment is comparable to structure: it determines how that potential can function and express itself.

4. What Did "Fertiliser-Responsive" Really Mean?

The Green Revolution produced important genetic changes in cereals. Semi-dwarf wheat and rice were selected for shorter stature, stronger standing ability and high grain production under abundant nutrient and water supply. Their architecture allowed them to carry more grain without lodging as readily as many taller varieties under high nitrogen conditions.

But the phrase "fertiliser-responsive" can mislead. The plant does not recognise a fertiliser company or a fertiliser bag. It responds to nutrient availability and to its physiological ability to acquire and use those nutrients.

Over time, two propositions became almost interchangeable in agricultural thinking: high nutrient availability and high synthetic fertiliser application. They are not scientifically identical. Synthetic fertiliser is one human method of increasing soluble nutrient supply. A biologically functioning soil uses a different pathway: mineral weathering, decomposition, microbial transformations, mycorrhizal associations, root exudation and continuous recycling.

5. Why Short Plants Were Important to ACI

Tall traditional wheat and rice varieties can respond strongly to nitrogen, but part of the additional growth may appear as stem and leaf biomass. Under high nitrogen application, excessive vegetative growth can increase lodging risk. Semi-dwarf architecture helped redirect more biomass toward grain while keeping the plant shorter and mechanically stable.

This characteristic was especially useful in the ACI environment, where roots were often constrained by repeated tillage, compaction and hardpan, while nutrients and water were supplied externally.

PQNK changes that environment. The one-time breaking of hardpan opens pathways for future roots and water. Once repeated disturbance stops and soil structure rebuilds, roots can explore the soil according to their genetics and the physical environment rather than being repeatedly confined by the production system.

6. PQNK Does Not Feed the Variety. It Restores the Feeding System.

ACI asks: what fertiliser dose should we apply to this variety?

PQNK asks: what conditions allow the plant, roots and soil biology to regulate their relationship naturally?

Under PQNK, the objective is moist but aerated soil, no repeated disturbance, permanent organic cover and encouraged biodiversity. Roots occupy the soil and release carbon compounds. Microorganisms participate in nutrient transformations. Mycorrhizal fungi extend the effective nutrient-acquisition network. Organic residues are continuously recycled.

This is why PQNK describes nutrition as demand-driven rather than bag-driven. The phrase does not mean that microorganisms behave like a computerised fertiliser dispenser. It means that nutrient acquisition is coupled to living roots, biological activity, water movement and plant demand rather than being governed primarily by periodic applications of concentrated soluble salts.

7. OP, Hybrid, HYV or GMO: Biology Still Governs the Plant

The basic soil principle does not change according to the label printed on the seed packet. An open-pollinated or heirloom variety, a modern high-yielding variety, a hybrid and a genetically modified variety all remain plants. Their genetics differ, but they all require water, oxygen, light, carbon dioxide, mineral nutrients, functioning roots and a biologically active environment.

PQNK does not manufacture a separate soil system for each category of seed. It restores the natural production environment in which the genetic potential of each plant can be expressed.

A hybrid can express hybrid vigour. A semi-dwarf HYV can express its high harvest index. An heirloom variety can express characteristics accumulated through generations of local adaptation. A GMO can express its engineered trait. But none escapes the biological laws governing plant life.

8. What Would a Green Revolution Variety Do Under Mature PQNK?

If a modern HYV is placed in mature PQNK soil, with deep rooting opportunity, moisture and aeration properly managed, active biological nutrient cycling and no repeated disturbance, there is no sound biological reason to assume that it suddenly loses its genetic yield potential merely because synthetic fertiliser is absent.

The useful research questions are therefore: how much of its potential can it express under PQNK? How does its root system change? How does yield compare? How does grain composition change? How much water and purchased input are required? How does the same genotype perform under ACI and PQNK?

These are measurement questions. They are more useful than assuming that “fertiliser-responsive” means “fertiliser-dependent.”

9. Native and Heirloom Seed Deserve Particular Attention

An heirloom or locally maintained open-pollinated variety is not merely an old genotype. It represents generations of interaction among genetics, local environment and human selection. Farmers repeatedly retained seed from plants that performed under local soils, temperatures, rainfall patterns, diseases, culinary preferences and cultural requirements.

Locally adapted germplasm is therefore both biological and cultural capital. Once a locally maintained variety is no longer grown and its seed is no longer conserved, useful genetic diversity can disappear.

Preserving natural, traditional, heirloom and farmer-maintained seed is not nostalgia. It is biological risk management. We should conserve it while it is still available.

10. Pollen Does Not Respect the Farmer’s Boundary

Saving traditional seed requires more than storing seed after harvest. In cross-pollinated crops, its genetic continuity must also be protected while the crop is flowering.

Maize is the clearest example. It is predominantly wind-pollinated and naturally outcrossing. Pollen released from a tassel can fertilise silks on other maize plants. If pollen from a GM maize plant fertilises a non-GM plant, the mother plant itself is not transformed. The newly formed kernel, however, is an offspring of both parents and can inherit the engineered trait carried by the pollen parent.

If such kernels are harvested and replanted as seed, the introduced gene can enter the farmer-maintained population. In following generations it can segregate, persist, spread further through crossing, or be removed through deliberate selection and genetic purity management.

This makes pollen-mediated gene flow fundamentally different from a one-season chemical exposure. Gene flow can become an inheritance event.

A field boundary is a physical and legal boundary. It is not necessarily a reproductive boundary.

11. Maize: Distance Reduces Gene Flow, but Does Not Create a Genetic Wall

Most maize pollen is deposited relatively close to the source, and the probability of successful cross-fertilisation generally falls rapidly with distance. But complete confinement cannot be assumed. USDA reviews report that most pollen moves only metres from the donor plant, while low-level gene flow has been detected hundreds of metres away

under some field conditions. Weather, wind direction, field size, flowering synchrony, pollen competition, barriers and the receiving variety all influence the result.

For that reason, KP-198 should not use “one mile” as a universal contamination radius. Long-distance pollen movement can occur, but pollen movement is not the same as successful fertilisation, and the frequency of gene flow normally becomes very low with increasing distance.

The practical conclusion is nevertheless important: a farmer growing non-GM open-pollinated maize near GM maize cannot assume that the field boundary alone guarantees genetic isolation. Seed conservation may require isolation distance, flowering-time separation, border management, controlled seed multiplication and, where genetic purity is critical, testing.

12. The Issue Is Generational

The importance of maize gene flow appears at the next planting.

A non-GM ear can contain kernels fathered by different pollen sources. If a GM pollen grain fertilises one ovule, the resulting kernel may carry the engineered trait even though neighbouring kernels on the same ear may not. When farmers save grain as seed, those kernels become the next generation.

Repeated pollen-mediated introgression can therefore alter the genetic composition of an open-pollinated population over time if incoming genes are repeatedly introduced and retained.

This is why saving seed means more than saving kernels. It means protecting genetic continuity from one generation to the next.

13. Maize Is Not the Only Crop, but Crops Differ Greatly

Pollen-mediated gene flow is not unique to maize. The risk, however, differs greatly among crops because reproductive biology differs.

Maize is particularly important because it is naturally cross-pollinated and wind-pollinated. Other crops may be predominantly self-pollinating, insect-pollinated, or capable of crossing with related cultivated or wild populations. Wheat, for example, is predominantly self-pollinating but measurable pollen-mediated gene flow has been documented under field conditions. Therefore the maize example should not be mechanically applied to every crop.

The governing principle is broader: the reproductive biology of each crop determines how readily genes can move beyond the field in which a variety was intentionally planted.

14. Seed Sovereignty Is Part of Food Sovereignty

Modern seed has increasingly become intellectual property. This does not apply exclusively to GMO varieties, and not every commercial seed operates under identical restrictions. But patents, contracts, licensing arrangements and concentrated ownership can limit the farmer’s freedom to reproduce particular commercial genetics.

Pollen-mediated gene flow adds another dimension. A farmer may choose not to plant a GM variety yet still need to manage incoming pollen if GM maize is flowering nearby. Seed sovereignty therefore requires not only access to seed but also practical space to reproduce chosen germplasm with acceptable genetic purity.

The concern is larger than genetic engineering itself. A food system that loses locally reproducible seed and becomes dependent on externally controlled reproductive material also loses part of its resilience.

Future farmers should inherit options rather than dependency. Seed conservation is insurance. Innovation can continue, but it should not require the disappearance of genetic resources and farmer-maintained alternatives.

15. GMO Genetics Must Be Separated From the ACI Production Package

It is not scientifically defensible to say that all GM crops inherently require huge quantities of water and fertiliser. Genetic modification is a breeding technology, and different GM varieties carry different traits.

The stronger PQNK criticism concerns the production architecture into which many commercial GM crops have been introduced. When seed is sold within a system built around monoculture, repeated disturbance, chemical weed control, external fertilisation, irrigation dependence and proprietary input packages, the farmer can become dependent upon the whole production architecture.

That dependency should not automatically be attributed to a single inserted gene. It belongs to the system.

Likewise, it is too broad to claim that GM seed, merely because it is genetically engineered, necessarily damages soil. Soil condition is shaped powerfully by the production system imposed upon it. Continuous disturbance, bare soil,

inundation, concentrated soluble inputs and reduced biological diversity can damage soil function regardless of the genetic label on the seed.

16. The PQNK Position on GMO Is More Fundamental Than "For" or "Against"

PQNK does not need to build its case on fear of a particular seed technology. Its questions are more fundamental: Who controls the seed? What genetic diversity are we losing? Under what production system is the seed being grown? Can farmers reproduce locally adapted seed? Can neighbouring farmers protect chosen seed populations from unwanted gene flow? Does the production system restore or degrade the living soil?

A genetically modified seed does not invalidate the natural laws governing agriculture. Neither does an heirloom seed automatically repair abused soil.

Put an heirloom seed into waterlogged, compacted, repeatedly tilled, biologically impoverished soil and its potential will also be suppressed. Put a modern variety into a functioning biological soil and it still responds as a plant.

PQNK therefore separates three questions that are often confused: the genetics of the seed, the ownership and reproductive control of the seed, and the production environment in which the seed is grown.

17. What PQNK Field Observations Are Telling Us About Food Quality

PQNK farmers have repeatedly reported differences in crops grown under the system, including stronger taste, perceived improvement in grain and produce quality, eating smaller quantities before feeling satisfied, and remaining without hunger for longer periods.

These observations matter: they are field evidence. But they should also be measured. Nutrient analysis can examine mineral and biochemical composition. Controlled sensory evaluation can examine taste. Human studies can examine satiety and metabolic response.

We should not infer a lower glycaemic index merely from reports of improved taste or prolonged satiety. Glycaemic index is a specific physiological measurement. If the same genotype grown under PQNK produces a different metabolic response, that should be demonstrated directly.

PQNK does not ask science to validate whether natural biological processes exist. It asks science to measure what happens when agricultural plants are returned to an environment governed more closely by those processes.

18. The Most Important Experiment Has Barely Been Asked

A revealing comparison would take several genetically different wheats or other crops: a locally maintained heirloom, an open-pollinated improved variety, a modern Green Revolution type and, where relevant and legally permitted, a genetically engineered variety.

Each genotype would be grown under two production systems. Under ACI it would receive the conventional technological package. Under a mature PQNK system it would grow in an undisturbed, mulched, biodiverse and moisture-managed living soil.

Measurements would include grain yield, total biomass, root depth and distribution, water use, nutrient acquisition, protein and mineral composition, grain quality, pest and disease behaviour, input cost, net return, taste and, where appropriate, human nutritional response.

Such an experiment would not determine whether nature works. It would reveal how different genetics express themselves under two radically different production environments.

19. The Genetic Potential Should Not Be Judged Inside a Degraded Environment

A recurring mistake in agriculture is to observe a plant inside an artificial or degraded production environment and attribute its behaviour entirely to genetics.

A shallow root system may partly reflect genetics. It may also reflect hardpan. Poor nutrient acquisition may partly reflect genetics. It may also reflect disrupted biological relationships. Drought stress may partly reflect climate. It may also reflect degraded soil structure and poor soil moisture management.

The texture-versus-structure principle provides the conceptual key: we should not judge the productive capability of clay merely from degraded, compacted clay. Biological structure can fundamentally change how that same mineral material functions.

Likewise, we should be cautious about judging the ultimate capability of a seed only by observing it inside ACI.

20. Save the Seed Before We Lose the Choice

PQNK strongly supports preservation of locally adapted, open-pollinated, heirloom and farmer-maintained seed. Not because every old variety is necessarily superior. Not because every modern variety is inferior. We preserve them because genetic diversity is accumulated biological capital. Once lost, it may be impossible to reconstruct.

For cross-pollinated crops such as maize, conservation must include reproductive stewardship. Seed stocks, isolation, flowering time and genetic purity matter because pollen can alter the genetics of the next generation.

A farmer should be able to compare local germplasm with modern germplasm under a functioning living-soil environment and make the choice on the basis of yield, resilience, quality, nutrition, taste, economics and independence.

The seed should serve the farmer and the food system. The farmer should not become merely the final licensed operator of somebody else's reproductive technology.

21. From Fertiliser-Responsive to Ecosystem-Responsive

Perhaps even the language needs changing. Instead of asking, "Is this a fertiliser-responsive variety?" we should ask: "How effectively can this genotype express its potential within a functioning biological production system?"

A seed is not a fertiliser-response machine. It is a living package of inherited potential. The soil is not merely an inert anchoring medium into which humans must pour the chemistry required to manufacture a crop.

The seed and the soil are both biological participants in a larger system.

The more useful concept is ecosystem responsiveness: how well can the plant use the water, minerals, air, sunlight, biological partnerships and physical rooting environment available to it when the production system is functioning properly?

Conclusion

The Green Revolution changed plant genetics and produced varieties capable of converting abundant resources into high grain yields. That achievement should be understood accurately. But it does not prove that synthetic fertiliser is biologically inseparable from those genetics.

PQNK takes that relationship one level deeper. Instead of continuously engineering the plant's environment through tillage, flooding, soluble fertiliser and chemical protection, PQNK seeks to restore the living production system around the seed.

The seed determines inherited potential. The living soil determines a major part of the environment in which that potential is expressed. Yet seed genetics are not frozen across generations. When sexually compatible plants cross-pollinate, the next generation can carry genes from both parents. In maize, this makes pollen-mediated gene flow a practical seed-sovereignty issue wherever GM and non-GM crops flower within reproductive reach of one another.

Humanity must therefore protect both the soil environment and the genetic inheritance accumulated in traditional seed. Save the natural seed while it still exists. Preserve farmer choice before diversity becomes dependency. Protect the ability to reproduce that seed with acceptable genetic purity.

The future of agriculture should not require choosing between science and nature, or between modern genetics and traditional genetics. The real task is to place genetics, old or new, back inside the production system that made plant life possible in the first place.

Key Propositions

- The seed carries inherited potential; the production environment governs its expression.
- "Fertiliser-responsive" does not mean "synthetic-fertiliser dependent."
- PQNK restores the soil environment rather than prescribing a separate feeding package for each variety.
- OP, heirloom, hybrid, HYV and GMO seeds all remain subject to the same biological laws of plant life.
- Traditional and locally adapted seed should be conserved as biological capital and as protection of farmer choice.
- In maize, pollen-mediated gene flow can introduce a GM trait into seed produced on a non-GM plant; if that seed is replanted, the introduced gene can enter subsequent generations.
- Distance reduces maize gene flow, but no single distance should be presented as a universal contamination radius.
- Seed sovereignty includes the practical ability to reproduce chosen germplasm with acceptable genetic purity.

- GMO genetics should be distinguished from the ACI production package and from proprietary seed-control systems.
- PQNK field observations on taste, satiety and food quality deserve systematic nutritional and metabolic measurement.
- The most revealing comparison is the same genotype under ACI and mature PQNK environments.

Scientific Note on Maize Pollen

USDA APHIS reviews of maize gene flow report that most maize pollen is deposited close to the donor plant, cross-fertilisation generally declines rapidly with distance, and low-level long-distance gene flow can occur under favourable conditions. Published USDA material also cautions that pollen dispersal is not equivalent to successful fertilisation. For this reason KP-198 deliberately avoids presenting one mile as a universal contamination radius.

Selected supporting sources:

- USDA APHIS. Draft Environmental Assessment 09-055-01p. Discussion of pollen-mediated gene flow and mitigation in corn.
- USDA APHIS. Response to Comments 07-152-01p. Discussion of corn pollen characteristics and declining gene flow with distance.
- USDA APHIS. Draft Environmental Assessment for DP23211 Maize (20-203-01p). Discussion of corn pollen dispersal and cross-pollination.
- USDA ARS. Research on pollen-mediated gene flow in winter wheat, demonstrating that crop reproductive biology strongly affects gene-flow risk.

- ACI: Ancient Conventional Industrial
- PQNK, To Be Pronounced As 'Picnic', Which Stands For Paedar Qudratti Nizam Kashatqari, And Means: The Regenerative & Sustainable Pristine Organic Farming System.